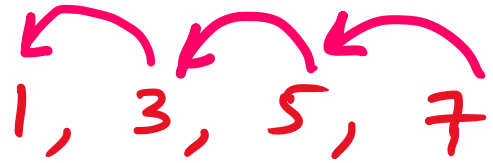


Arithmetic Progression [AP series]

1, 3, 5, 7



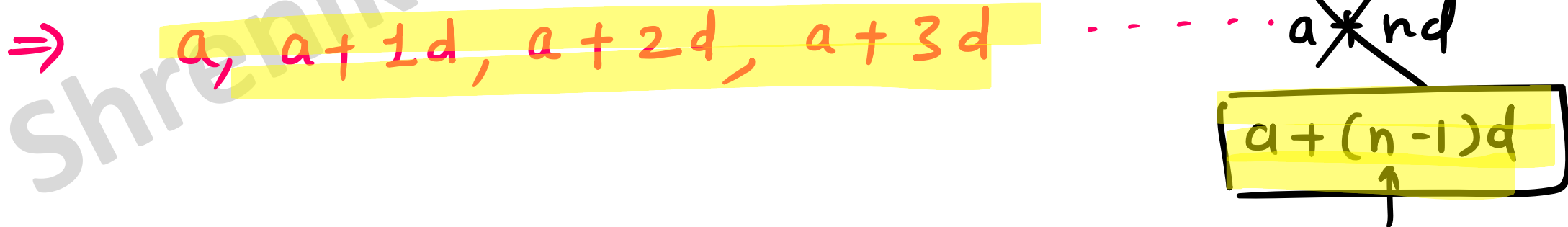
1, 1+2, 1+4, 1+6

1, 1+1×2, 1+2×2, 1+2×3

$d = \text{Next} - \text{prev}$

⇒ $a, a+1d, a+2d, a+3d, \dots, a+nd$

$a + (n-1)d$



AP series

① $a, a+d, a+2d, \dots, a+(n-1)d$

[n terms]

$d = \text{common difference}$

② Last term of AP series

$$T_n = a + (n-1)d$$

↑

Q]

1, 3, 5, 7, 9, 11

Sum of series.

Solⁿ:

(M1)

$$S = 1 + 3 + 5 + 7 + 9 + 11 = 36 //$$

(M2)

[AP series]

$$S = \frac{n}{2} [2a + (n-1)d] = \frac{6}{2} [2(1) + 5(2)] = 36 //$$

(M3)

$$S = \frac{n}{2} [a + T_n] = \frac{6}{2} [1 + 11] = 36 //$$

$$S = \frac{n}{2} [2a + (n-1)d]$$

$\downarrow$
 $a + a$

$$S = \frac{n}{2} [a + T_n]$$

Q] What is the **number of terms** in the series

■ 117, 120, 123, 126, 333

Solⁿ : $d = 3 \rightarrow \text{[AP series]}$

$$a = 117$$

$$T_n = 333$$

$$\Rightarrow T_n = a + (n-1)d$$

$$333 = 117 + (n-1)3$$

$$\Rightarrow n = 73 //$$

Q] If the number $n-3$, $4n-2$, $5n+1$ are in AP then what is the value of n ?

Concept:

a, b, c [AP]
 $\smile \quad \smile$

$$\Rightarrow b - a = d$$

$$\Rightarrow c - b = d$$

$$\Rightarrow b - a = c - b$$

$$2b = a + c$$

$$\Rightarrow \boxed{b = \frac{a+c}{2}}$$

Arithmetic
mean

$n-3, 4n-2, 5n+1$ are in AP

$$4n-2 = \frac{(n-3) + (5n+1)}{2}$$

$$8n-4 = 6n-2$$

$$\boxed{n=1}$$

Q] If **sum** of 1st 10 terms and 1st 20 terms of an AP are 120 and 440 respectively then what is the common difference?

80/n : $d = ?$ [AP series]

$$S = \frac{n}{2} [2a + (n-1)d]$$

$$\overset{120}{S_{10}} = \frac{10}{2} [2a + (10-1)d]$$

$$\therefore \boxed{24 = 2a + 9d}$$

$$440 = S_{20} = \frac{20}{2} [2a + (20-1)d]$$

$$\therefore \boxed{44 = \cancel{2a} + 19d}$$

②

$$\boxed{24 = \cancel{2a} + 9d}$$

$$d = 2$$

Q] If sum of 5 consecutive terms in AP is found to be 120, find the middle term

Soln: [AP series]

$$\Rightarrow a, a+d, a+2d, a+3d, a+4d$$

(M1)

$$S_5 = 120$$

$$a + a + d + a + 2d + a + 3d + a + 4d = 120$$

$$5a + 10d = 120$$

$$\Rightarrow a + 2d = 24 \quad \{\text{middle No.}\}$$

Q] If sum of 5 consecutive terms in AP is found to be 120, find the middle term

5 consecutive terms in AP

$$\cancel{a-2d}, \cancel{a-d}, a, \cancel{a+d}, \cancel{a+2d}$$

$$\sum a = 120$$

$$\boxed{a = 24}$$

Q] What will be the maximum sum of 44, 42, 40, ...?

(A) 502

(B) 504

(C) 506

(D) 500

[GATE 2013 CS]

44, 42, 40, 38, ... 4, 2

⇒ 2, 4, ... 44

⇒ AP series

$$\Rightarrow S = \frac{n}{2} [a + T_n] = \frac{22}{2} [2 + 44] = 506$$

$$\Rightarrow T_n = a + (n-1)d \Rightarrow 44 = 2 + (n-1)2 \Rightarrow n = 22$$

Q] The sum of eight consecutive odd numbers is 656.

The average of four consecutive even numbers is 87.

What is the sum of the smallest odd numbers and second largest even number?

[GATE 2014 EC]

Soln :

① [AP series]

$$d = 2$$

$$n = 8$$

$$S_8 = 656$$

$$656$$

$$S_8 = \frac{8}{2} [2a + (8-1)2]$$

$$a = 75$$

(smallest odd no.)

75, 77, 79, ... 89

②

$$\text{Avg} = \frac{\text{Sum}}{\text{total}}$$

$$\therefore \text{Sum} = \text{Avg} \times \text{total}$$

$$\frac{4}{2} [2a + 3(2)] = 87 \times 4$$

$$\Rightarrow \boxed{a = 84}$$

$$n = 4$$

$$d = 2$$

$$S_4 = \frac{n}{2} [2a + (n-1)d]$$

$$84, 86, 88, 90$$

[2nd largest]

$$\text{Ans} = 75 + 88$$

$$= 163 //$$

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Q] In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

[GATE 2014 EC]

$a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, \dots, a_{12}$

①

$a_1, a_2, a_3, a_4, a_5, a_6, a_7$

$a_8, \dots, a_{12}$

$$S_{12} - S_7 = \text{Req}$$

$$S_{12} - S_7 = \text{Req}$$

$$S_5 = 425 \quad [AP \Rightarrow d=2]$$

$$\frac{5}{2} [2a + (5-1)2] = 425$$

$$\Rightarrow \boxed{a = 81}$$

$$S_{12} - S_7 = \text{Req}$$

$$a = 81$$

$$d = 2$$

$$\frac{12}{2} [2 \times 81 + (12-1)2]$$

—

$$\frac{7}{2} [2 \times 81 + (7-1)2]$$

$$= 495 //$$

Q] What is the average of all multiples of 10 from 2 to 198?

(A) 90

(B) 100

(C) 110

(D) 120

[GATE 2014 CS]

$$\textcircled{1} \quad \text{Avg} = \frac{\Sigma}{n} = \frac{1900}{19} = 100 //$$

$$\textcircled{2} \quad \{ \textcircled{10}, 20, 30, \dots, \textcircled{190} \}$$

• AP series

• $d = 10$

• $S = \frac{n}{2} [10 + 190]$
 $= 1900$

$$T_n = a + (n-1)d$$

Geometric Progression [G.P]

2, 4, 8, 16, 32

$$\frac{4}{2} = \frac{8}{4} = \frac{16}{8} = \frac{32}{16} = 2$$

$r =$

[Common ratio]



[G.P]

$$a, ar, ar^2, ar^3, \dots, ar^{n-1}$$

G.P series 'n terms'

$$T_h = ar^{h-1}$$

Shrenik Jain – Study Simplified

$$\therefore S = a + \cancel{ar} + \cancel{ar^2} + \cancel{ar^3} + \dots + \cancel{ar^{n-1}} + 0$$

$$\textcircled{-} \quad rS = \quad \downarrow \quad \cancel{ar} + \cancel{ar^2} + \cancel{ar^3} + \dots + \cancel{ar^{n-1}} + ar^n$$

$$S - rS = a - ar^n$$

$$S[1-r] = a[1-r^n]$$

$$S = \frac{a[1-r^n]}{[1-r]}$$

Sum of n terms of GP series

$$S = \frac{a [1 - r^n]}{[1 - r]}$$

Sum of ∞ terms of GP series

$$S = \frac{a [1 - r^n]}{[1 - r]}$$

r^∞

$$\left(\frac{1}{2}\right)^\infty = \frac{1^\infty}{2^\infty} = \frac{1}{2^\infty} = \frac{1}{\infty} = 0$$

$$(10)^\infty = \infty$$

$$S_{\infty} = \begin{cases} \infty, & r > 1 \\ \frac{a}{1-r}, & |r| < 1 \end{cases}$$

$$S = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

$$r=1$$

$$= \underbrace{a + a + a + a + \dots + a}_{n \text{ terms}}$$

$$S = a \times n = na$$

$$S_{\infty} = \infty \cdot a = \infty //$$